Somatosensory

We can think that vision is the main modality conveying information, but when we talk about all the different sensor modalities, we can imagine that all can be used for several tasks including aspects of processing. The somatosensory modality is the first modality that develops in the human brain, you can consider touch as a very basic aspect of perception.

When we talk about somatic sensibilities we should imagine that we have a variety of receptors that are distributed throughout the body that provide different pieces of information about not just what we typically call touch, that is something (at least in his active form) that we use to recognize size, shape, etc…; we have also proprioception, the sense of static position and movement of limbs and the body, something that is not just related to perceiving the effect of the movement of our levers but also to represent body per se. All elements that should be embodied, we have to think also of the perception of our body.

The bodily senses

Somatic sensibility arises from information provided by a variety of receptors distributed throughout the body. Somatic sensibility has four major modalities:

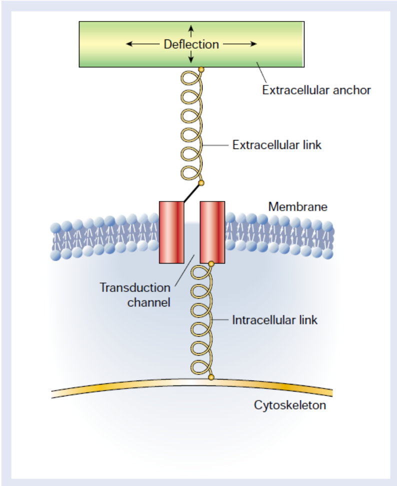
- discriminative touch (required to recognize the size, shape, and texture of objects and their movement across the skin).
- proprioception (the sense of static position and movement of the limbs and body).
- nociception (the signalling of tissue damage or chemical irritation, typically perceived as pain or itch), not only burning and stinging, but also chemical responses of the nerves to specific notch stimuli.
- temperature sense (warmth and cold).

These elements that are going to discriminate all this information relies on different population of receptors, typically capturing mechanic changings in mechanic forces, that are not just located on the skin but also located in the muscles, in the joints, etc.. .All this piece of information from the periphery of the body are carried to the brain by different pathways, that run parallely inside the spinal cord, then in the brain stem and then in the thalamus, reaching the primary somatosensory cortex in the post-central gyrus of the parietal lobe.

When we think about touch we primarily relies on the idea of skin, that have different components, including glands that produce liquids not only for regulating temperature but also to favour the interaction of the skin with objects. We divide the receptors on the skin in three groups:

1. Mechanoreceptors, able to capture mechanical forces.
2. Nociceptors, able to capture painful stimuli.
3. Thermoreceptors, able to find thermal information.

This division of function is integrated also by a division on the morphology, with receptors on the body surface could be free type or encapsulated type. Despite their variety, all these receptors are actually able to capture every piece of information getting to the skin, and then affecting the permeability of the membrane, allowing the sensory neuron to transduce the signal to the spinal cord and then to the brain.



The membrane gets typically anchored with the protein to the cytoskeleton structure, and then there are some kind of information getting and moving in the extracellular anchor, and once this is deflected than the channel opens, inducing a sodium depolarization. This is how typically we transduce mechanical forces.

All these kinds of receptors are then associated to specific fivers. The AB fivers are large and myelinic, meaning a higher conduction velocity, maintained by the myelin around the nerve. The non-encapsulated (free nerve endings), instead rely on C fivers or Ad fivers; these receptors appear to be very important for primarily conveying information about nociceptive and painful stimuli and thermal changes, but these C and Ad fivers are each amyelinated and poorly myelinated, so with a slower conductivity.

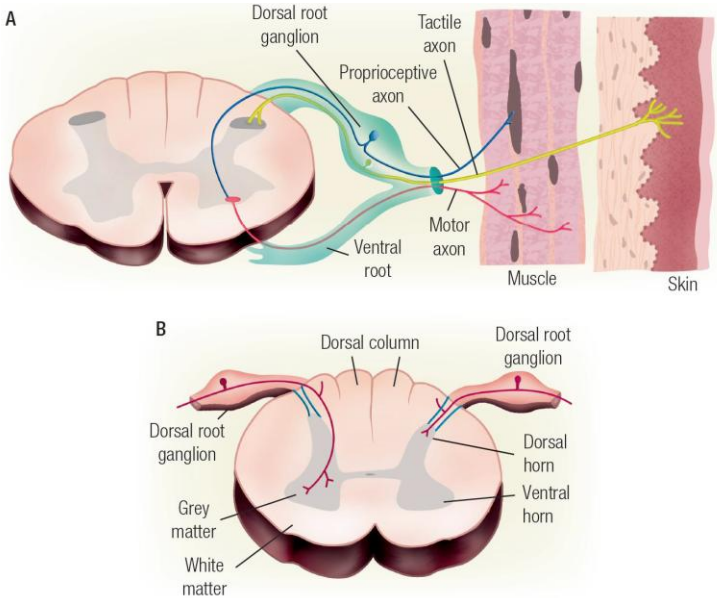
Table 22-1 Receptor Types Active in Somatic Sensation

Receptor type	Fiber groups[redacted]	Fiber name[redacted]	Modality
Cutaneous and subcutaneous mechanoreceptors			Touch
Meissner's corpuscle	Aα,β	RA	Stroking, fluttering
Merkel disk receptor	Aα,β	SAI	Pressure, texture
Pacinian corpuscle[redacted]	Aα,β	PC	Vibration
Ruffini ending	Aα,β	SAII	Skin stretch
Hair-tylotrich, hair-guard	Aα,β	G1, G2	Stroking, fluttering
Hair-down	Aδ	D	Light stroking
Field	Aα,β	F	Skin stretch
Thermal receptors			Temperature
Cool receptors	Aδ	III	Skin cooling (25°C)
Warm receptors	C	IV	Skin warming (41°C)
Heat nociceptors	Aδ	III	Hot temperatures (>45°C)
Cold nociceptors	C	IV	Cold temperatures (<5°C)
Nociceptors			Pain
Mechanical	Aδ	III	Sharp, pricking pain
Thermal-mechanical	Aδ	III	Burning pain
Thermal-mechanical	C	IV	Freezing pain
Polymodal	C	IV	Slow, burning pain
Muscle and skeletal mechanoreceptors			Limb proprioception
Muscle spindle primary	Aα	Ia	Muscle length and speed
Muscle spindle secondary	Aβ	II	Muscle stretch
Golgi tendon organ	Aα	Ib	Muscle contraction
Joint capsule mechanoreceptors	Aβ	II	Joint angle
Stretch-sensitive free endings	Aδ	III	Excess stretch or force

[1] See [redacted].[2] Pacinian corpuscles are also located in the mesentery, between layers of muscle, and on interosseous membranes.

These receptors at the level of the skin get back to the ganglia neurons, located dorsally closed to the vertebra, so once these neurons get information from the periphery they then project it to the spinal cord, so on the periphery there is a long peripheral branch that gets information from the skin, transmitting the action potential to the soma and the to the synapsis on the spinal cord. These neurons are called T-shaped, because they have two axons, one is a peripheral axon that gets to the skin and the internal organs, and the other goes towards the spinal cord. These neurons are primarily located along the vertebra, because closer to the vertebra there is an empty channel where the spinal cord stays and information from the periphery has to get there.

The Bell-Megendie Law says that the anterior portion of the grey matter in the spinal cord (ventral horn) is primarily conveying motor information to the muscles, and dorsal horn receive sensory afference from the skin and the different body parts.



Once the information get from the skin to the dorsal ganglion route than it enters the spinal cord and goes up to the brain stem and then to the thalamus, and finally to the post-central gyrus in the cortex. In the spinal cord there is a fine organization of all the piece of information getting up and here all these pathways (thermal, proprioceptive, tactile) flow parallel up to the brain, and also here there is a specific organization that mirrors topography of the body. In the spinal cord there is a primary organization of the information that goes up to the brain.

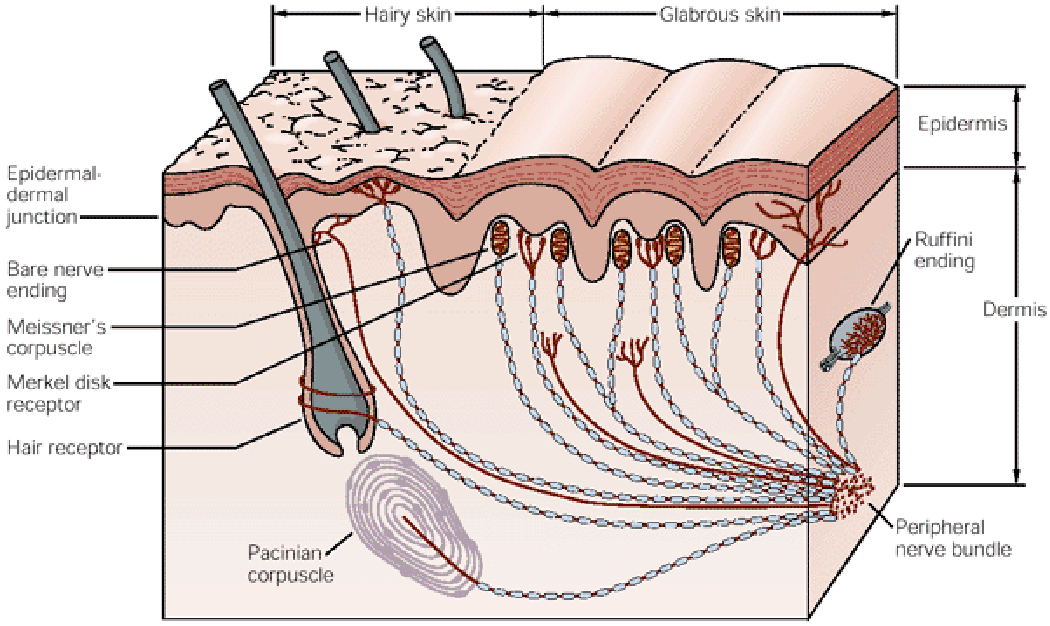
There are four major types of encapsulated mechanoreceptors, primarily specialized in providing information and capturing all mechanical information (pressure, vibration, tension), and all these receptors are collectively referred as high sensibility mechanoreceptors, because even weak mechanical stimulation of the skin induces them to produce action potentials. One aspect very typical of all sensor modalities. All low-threshold mechanoreceptors are innervated by relatively large, myelinated axons (type Aβ), ensuring the rapid central transmission of tactile information.

There are two attributes that refer to somatic sensations: epicritic and protopathic.

Epicritic sensations involve fine aspects of touch and are mediated by encapsulated receptors to: detect gentle contact of the skin and localize the position that is touched (topognosis); discern vibration and its frequency/amplitude; resolve by touch spatial detail, such as the texture of surfaces, and the spacing of two points; recognize the shape of objects grasped in the hand (stereognosis). You should consider tactical information as information that needs to be elaborated, because in this case we don’t immediately have the whole information (like in vision) but we have a sequential acquisition of information that we have to merge.

Protopathic sensations involve pain and temperature senses (as well as itch and tickle) and are mediated by receptors with bare nerve endings.

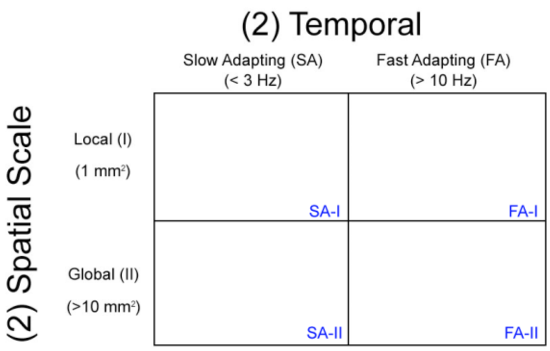
If we want to know where these high sensitivity receptors are located it's very difficult.



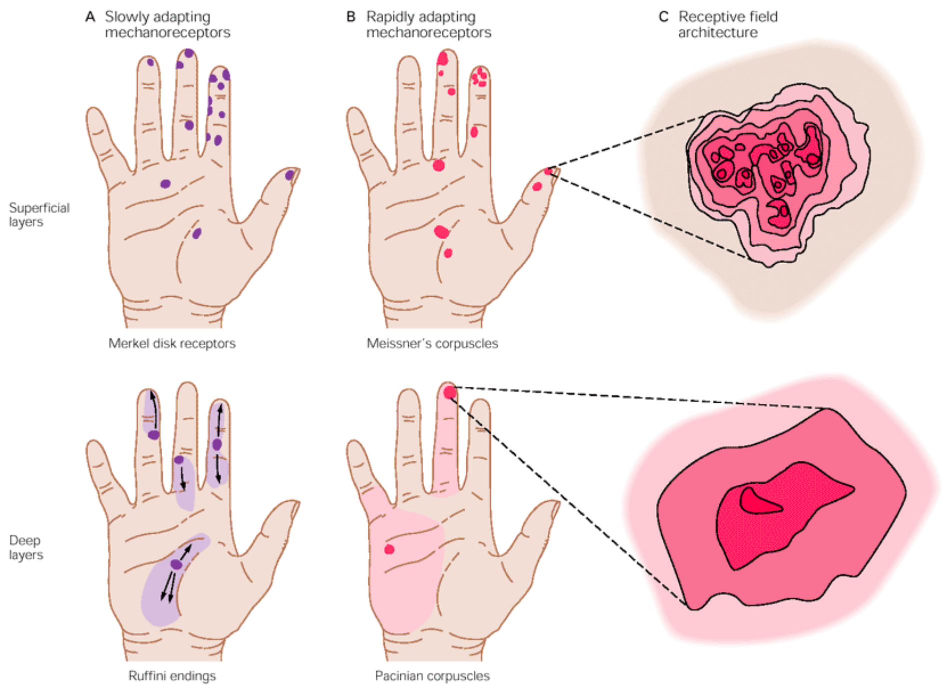
The glabrous skin is made up of ridges that are actually able to better capture and discriminate texture, so they favour the capture of very small mechanical forces in the skin. Some encapsulated receptors are located immediately subcutaneously, where the ridges are, while others are encapsulated deeper, so that we distinguish cutaneous and subcutaneous receptors. In addition to the four encapsulated receptors, we also have free nerve endings that reach the surface of the skin, but there are also nerve endings getting around the hairs or directly on the hairy surface.

There is a 2 by 2 design, and an important aspect is the location, the Messner and the Merkel are located cutaneously, the Pacinian and the Ruffinian are located subcutaneously. But this design is not only related to location, but also to the kind of response. In order to define the time of response, receptors may act on two different ways: they can discharge at the beginning and then slowly adapt so the firing rate is decreasing at constant for all the time it takes to stimulate; or you can have rapidly adaptive responses and receptors that discharge at the beginning and at the end of the stimulus. The Messner and the Pacinian corpuscles are rapidly adapting and the other two are slowly adapting.

Talking about tactual modality, we are actually saying that all our body is spread by different kind of receptors that capture several pieces of information and energy that get into the body.



Each receptor has his own receptive field, and the deep layers layers receptors (like the Ruffini) have a broader one. Typically, the Ruffini is involved in monitoring and capturing movements over the skin, instead the receptive field architecture of superficial receptors offers a higher resolution than the deeper ones. This aspect of resolution is very important, because is determined by two elements, one is the individual receptive field, and the other is the density.



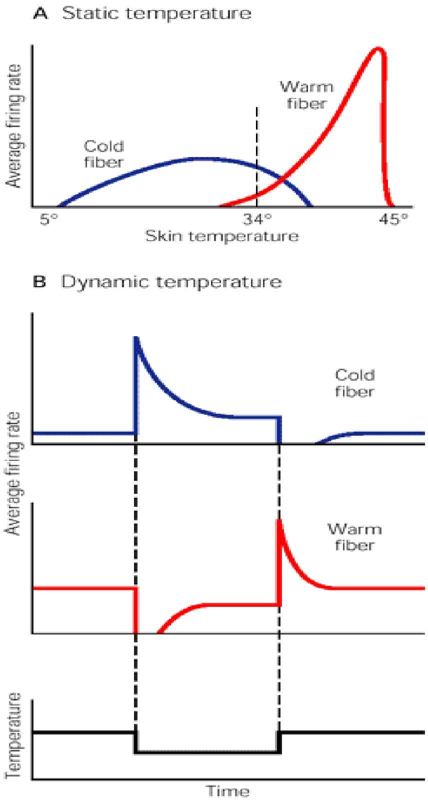
The deeper ones, like the Ruffini, are more present in the palm of the hand, while the others are primarily located in the fingertips, and this is where we rely for discriminating elements. It's important to distinguish areas that have high density of receptors or high resolution of discrimination. You can improve the experience of discriminating but not the physiological number of receptors.

Other parts of the body may not have any utility in discriminating objects or texture, so typically discrimination of two points is almost up to 4-5 cm, so two stimuli in our leg are not two points but just one is very close.

One thing that is also important to know is that each family of receptors may have a specificity in response, if you want to study different receptors you can just change the frequency of the vibro-tactile stimuli to probe response from different receptors. Touch detection thresholds vary with vibrotactile stimulus frequency and probe area consistent with the 4-channel model.

One of the things that we have to keep in mind is the population code, hardly we have one receptor activated when a stimulus arrives, most of the time we have several receptors working and responding at the same time, and the brain analyse it with what is called population code. It means that each receptor provides a piece of information and the brain has to merge it into a unique representation and synthesis.

If we go to observe other information, the very specific information we get to the skin is thermal perception, the other types of information (texture, shape) can be seen also by other perception modalities. Whether an object is cold or hot you have to touch it. When we talk about thermal sensation, the discrimination of the temperature is the difference between the air temperature



that we sense on our skin and the temperature of the object. Our thermal receptors work in a range from 5 to 45 °C, below or above there is no discrimination, only pain and damage. When we look at temperature we have two fivers, cold fivers and warm fivers, and these also act in two different ways. The cold fivers are increasing and going up to stimulate from 5 °C to a peak and then going down until body temperature (37-38 °C). Thermal receptors modulate the firing as a function of temperature. The warm fivers start responding around 30-32 °C and they go up exponentially with their firing rate until they reach 45 °C and then their response drop.

Around body temperature the discrimination abilities are much higher, because there are two fivers responding to the stimulus.

Proprioception

Proprioception is the sense of position and movement of our body. We acquire two pieces of information, the stationary position of our body and limbs (actually related to body representation and awareness), and the sense of limb movement (kinaesthesia). There are many disorders related to this perception modality, there are problems in how our brain represent and is aware of our body parts.

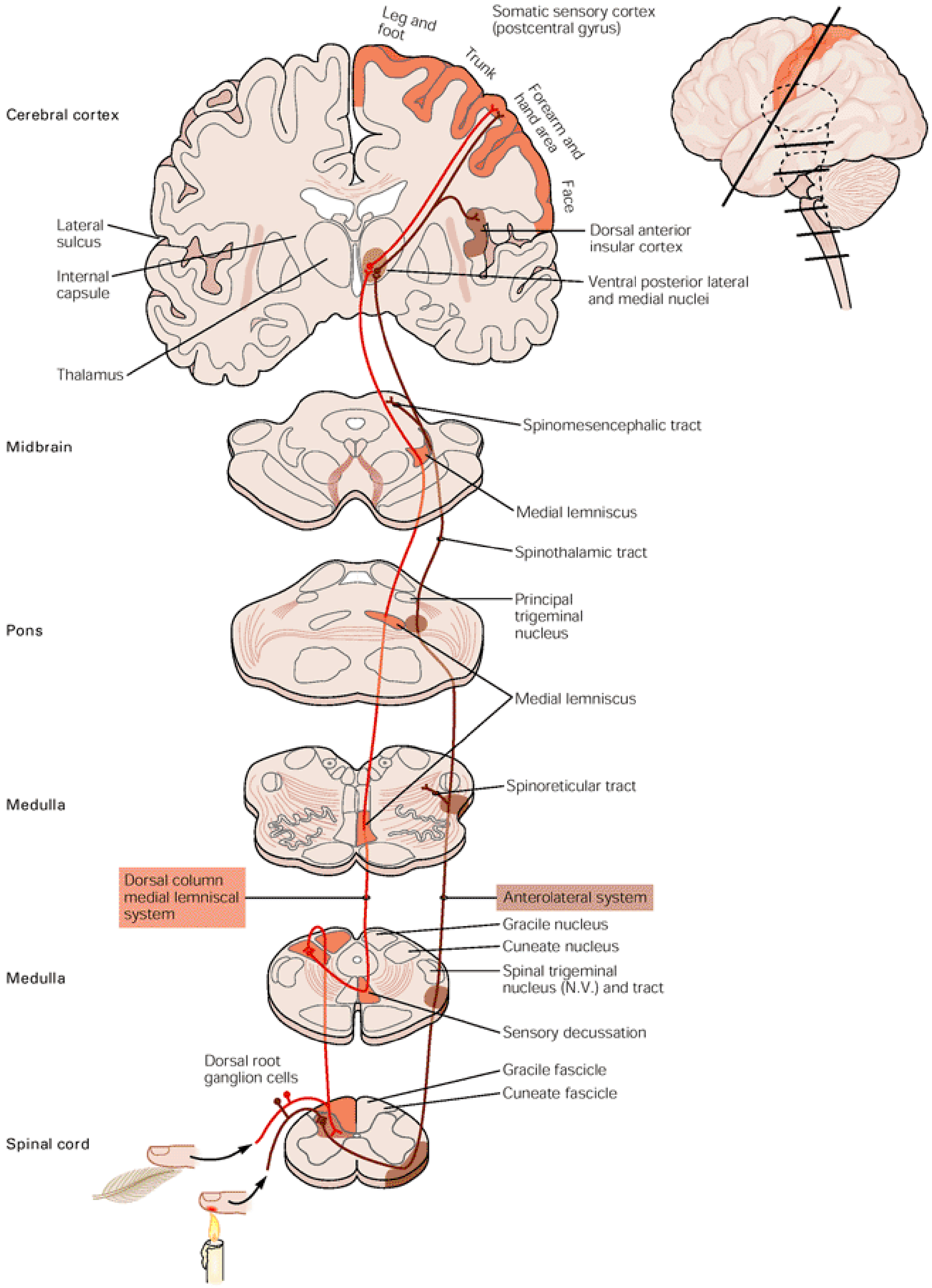
It relies in three families of receptors that are located in muscles and joints and are able to signal the spin and direction of the limb. These receptors are also located in the tendon or in the joint capsule, and they can sense the tension of the tissue and the flexion and extension of joints and muscles.

The body is separated in dermatomes, that reflect the surface of the skin that is innervated by each ganglion route (takes the name of the vertebra). Each of these areas corresponds specifically to the neurons located on a specific ganglion structure. Depending on the location of the pain we are actually able to understand where the problem is at level of the ganglion routes, since each of the part of the skin is innervated. (search for examples)

Once the information enters the posterior horn, the main pathway of touch and proprioception is what is called the dorsal column. The information gets in the posterior horn of the spinal cord and the axon that enters progress ipsilaterally up to the brain stem and only at the level of the medulla it crosses the spinal cord and goes up to the thalamus and then to the specific location in the post-central gyrus.

Also, what else tell us about the topographic organization is that axons that enter the cord in the sacral region are found near the midline of the dorsal columns; axons that enter the cord at successively higher levels are added in progressively more lateral positions. Even in the spinal cord we have very specific topological organization.

So, the tactual component is conveying information on the hemi body ipsilaterally in the spinal cord ant then only at the level of the brain stem it crosses to get to the other side. Information for pain stimulation are immediately crossing the spinal cord and they convey information up in the spinal cord contralaterally, to the hemi body that they are actually monitoring.

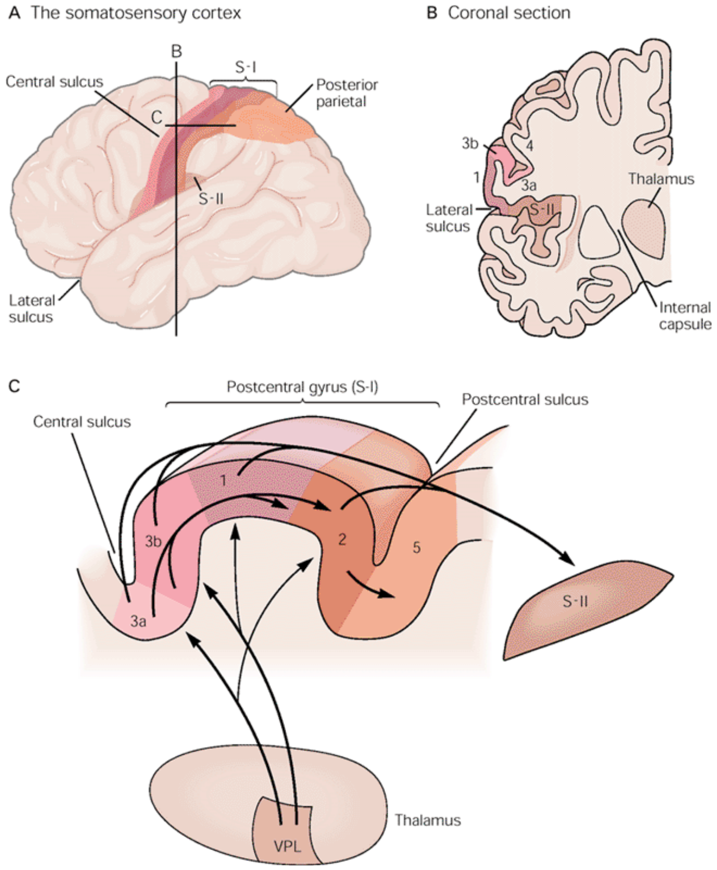


The other aspect is that the somatotopic organization of the axons relaying input from receptors in the skin is maintained throughout the entire ascending somatosensory pathway, through the thalamus, to the somatosensory areas in the postcentral gyrus of the cerebral cortex.

The ability to recognize objects placed in the hand on the basis of touch alone is one of the most important and complex functions of the somatosensory system: *stereognosis*. Many familiar objects are much larger than the receptive field of any one receptor in the hand. In addition, objects excite more than one kind of receptor.

The brain organization is not just random, shuffled pathways that then get ordered at level of the cortex, but is already ordered immediately from the spinal cord, up to the thalamus. In the spinal cord the fivers that you find close to the midline are the sacral ones, and as soon as you go up and more fivers enter, they go more laterally, so it is already organized.

Once the information gets to the thalamus it then reaches the post-central gyrus, in which there are areas (3a, 3b, 1, 2 ,5), looking at the coronal section.



These portion of the cortex (3a, 3b) are responding to tactile and proprioception, and then this information is merged together as soon as we move posteriorly in the brain, so areas 1 and 2 are already merging information from more higher complexity and when it gets to the posterior parietal cortex, we have integrated information between touch and proprioception. Together with the primary somatosensory area we typically describe the second somatosensory area (S-II), and these together are also getting information from other sensory modalities or subcortical structures, and areas 5 and 7 are merging information not just from the somatosensory cortex but also information that gets dorsally. Furthermore, as soon as we move from area 3a and 3b posteriorly, the information of a small part of the hand is merged into a broader area, and this is a typical phenomenon that we are going to see also in the visual cortex: at early stages the sensory pathway have a small receptive field, but once the information is integrated across different steps the receptive field gets broader.

The organization in columns is the same kind of organization that will be seen in the primary sensory visual area. Each chunk of the cortex is made up of six layers, and all neurons that are contained in a single column are processing the same kind of information, and this is related to a rapidly adapting receptor or slow adapting receptor. So, each column of the sensory cortex specifically contains information from one family of receptors and a specific function. There is different scale of functional organization across these columns, and the principle that rules the organization are: the functional response of mechanoceptors; the somatotopic organization of the cortex. If I have an information that gets only in the second finger, only his portion of the cortex will light up.

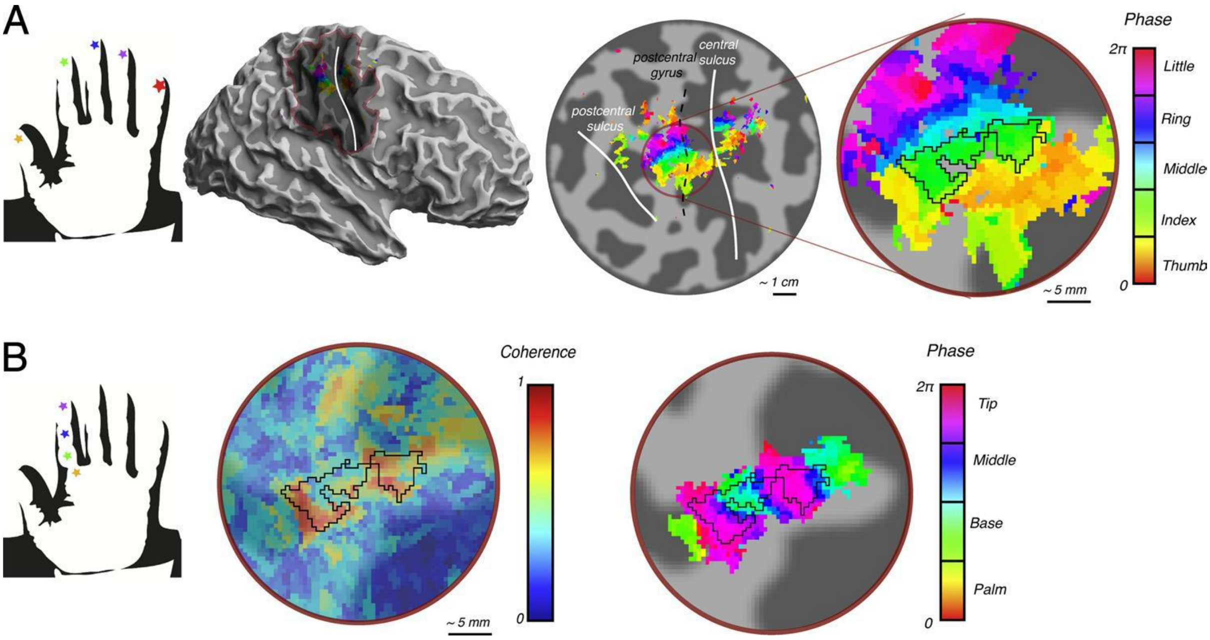
There are already mechanisms at the level of the skin that enhance specific information, the periphery of the sensory organs is already organized in order to provide specific way of discriminating different stimuli.

Haptics

Tactile discrimination normally entails active exploration. Psychophysical evidence indicates that relative movement between the skin and a surface is the single most important requirement for accurate discrimination of texture. Animal experiments confirm the dependence of tactile discrimination on active exploration.

Active touching, which is called ***haptics***, involves the interpretation of complex spatiotemporal patterns of stimuli that are likely to activate many classes of mechanoreceptors. Haptics also requires dynamic interactions between motor and sensory signals (tactile and proprioceptive), which presumably induce sensory responses in central neurons that differ from the responses of the same cells during passive stimulation of the skin. Proprioceptive signals because you have to have clear in mind where your hand is.

Tactual discrimination per se is something that occurs passively, when you have an active exploration and you have to merge several pieces of information (touch, proprioception, motor), this is what entails active exploration, and you call that haptics.



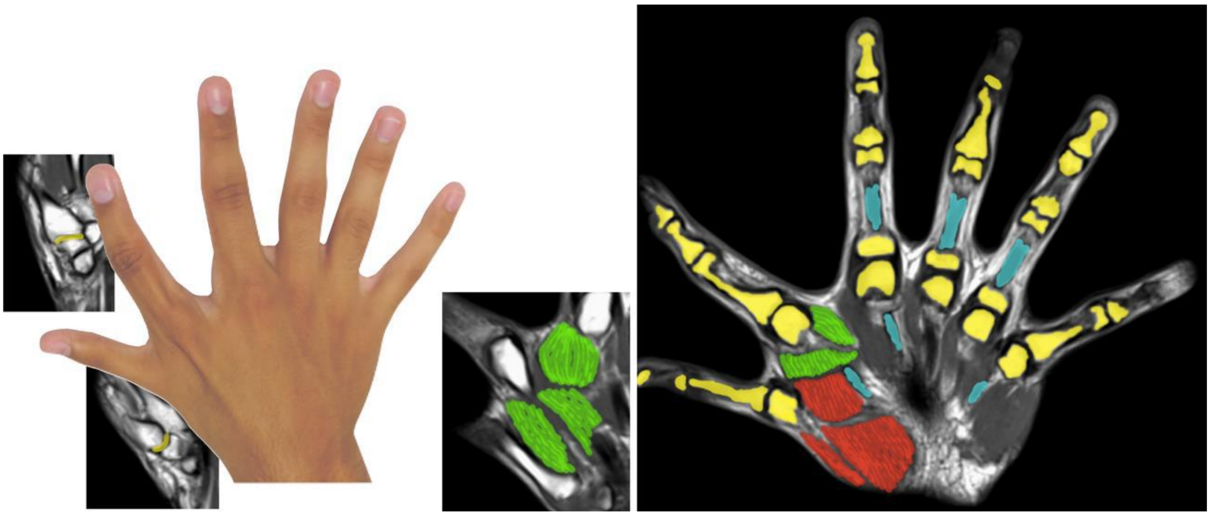
You can discriminate between the different fingers, but also between fingertips or regions closer to the palm.

To study the phenomenon, we can also use data acquired with electrocorticography, in which we have electrodes implanted on the cortical surface and the movie nicely tells how we process tactile information, which gets in the post-central gyrus and the post-central sulcus. Information that gets there is immediately spread to the motor cortex, so information from the tactile modality is mirrored in the motor cortex and then is slowly progressively sent backwards to the posterior parietal areas and then to the secondary tactile regions.

With brain imaging we can study different elements of tactile representation, but in some specific environments is not really easy, for instance MRI. Also, when you do haptic studies and you ask the subject to actively move and explore objects most of the time, he also moves his head, and this doesn't get along with MRI.

It's really difficult to distinguish the contribution that is related to touch from that related to motor, because of the tight coupling between motor processing and tactile information processing.

There are genetic disorders that can increase the number of fingers, and you still maintain the five finger organization, but you are also able to adapt a sixth finger representation in the cortex of the subject (augmented manipulation ability).



In the case of amputees, you have to cut the nerve, and the left other fivers detect the missing hand, so subject may have sensation about the limb that they don't have anymore, also with pain.

Once that you play with this kind of extra limb in the body and provide a new route of tactile stimulation, then you can map at brain level how this information is actually re-organized.

Pain and nociception

Pain is something that could be described in different ways, somehow is very distinctive among the different sensor modalities, and unlike other somatic and sensor modalities the quality of information that is conveyed by pain is less specific but has the characteristics of urgence. It is a sensory modality that primarily has to alert our brain, and this kind of alertness describes how the brain is actually the site that captures these elements of urgence.

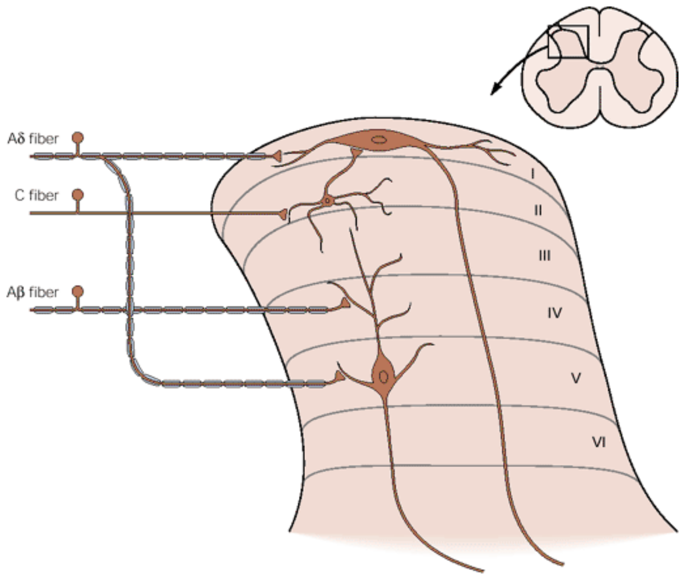
The sensations we call pain are the most distinctive of all the sensory modalities. Pain is a sub modality of somatic sensation and serves an important protective function: It warns of injury that should be avoided or treated. Pain has an urgent and primitive quality, a quality responsible for the affective and emotional aspect (I am able to get aware of this emotional aspect later) of pain perception. Moreover, the intensity with which pain is felt is affected by surrounding conditions, and the same stimulus can produce different responses in different individuals under similar conditions. The ability to reduce the perception of pain is called analgesia, but we can also exaggerate it or anticipate it.

That is why we separate pain itself from the perception of it.

Pain is a percept; it is an unpleasant sensory and emotional experience associated with actual or potential tissue damage. Although pain is mediated by the nervous system, a distinction between pain and the neural mechanisms of nociception - the response to perceived or actual tissue damage - is important both clinically and experimentally. Certain tissues have specialized sensory receptors, called *nociceptors*, that are activated by noxious insults to peripheral tissues. Nociception does not necessarily lead to the experience of pain.

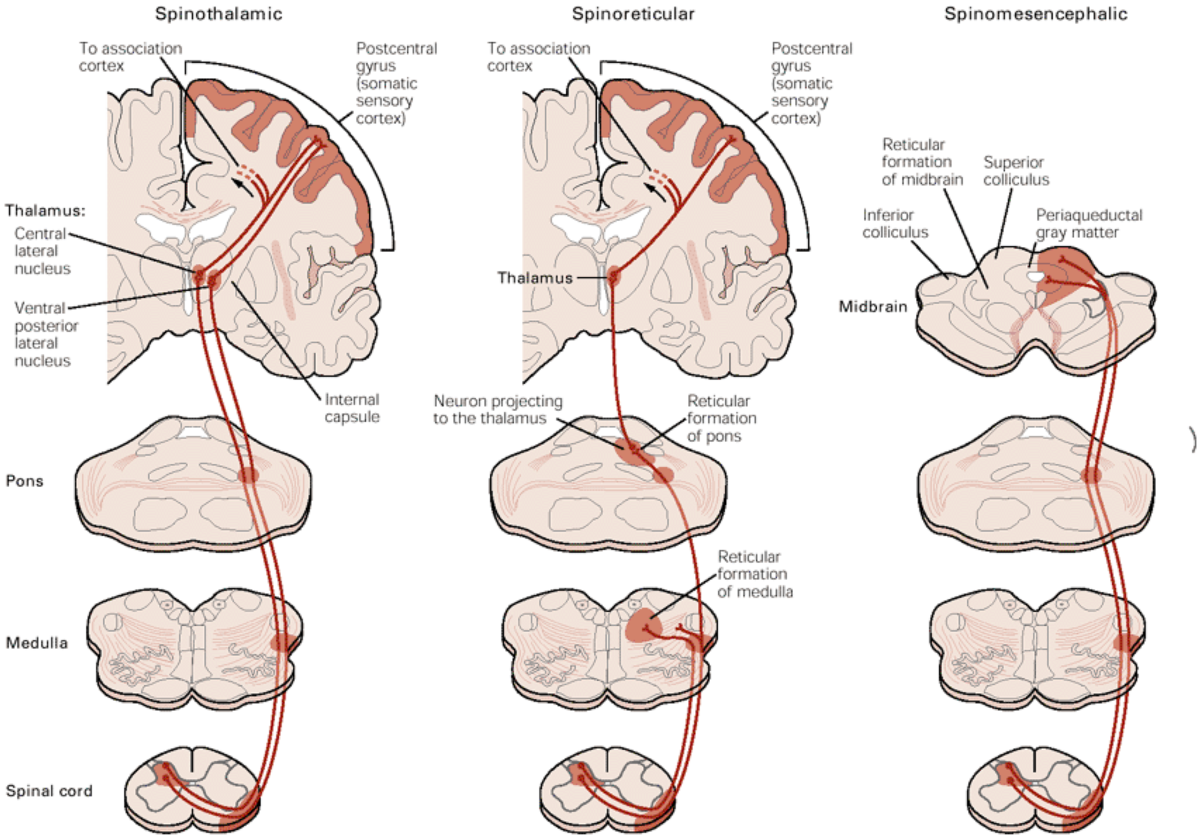
There are thermal nociceptors, mechanical nociceptors, polymodal nociceptors, that respond to different stimuli (mechanical, thermal) but are innervated by small diameters no-myelinated C fivers (velocity of less than 1 m/s).

Having two fivers that convey information from thermal and mechanical nociceptors and polymodal, allow to get generally what is called two ways of pain. These fivers are getting this idea of the first very brief input of pain and then the second longer wave of pain, you can notice it when you cut yourself, an immediate pain but it also continues later.



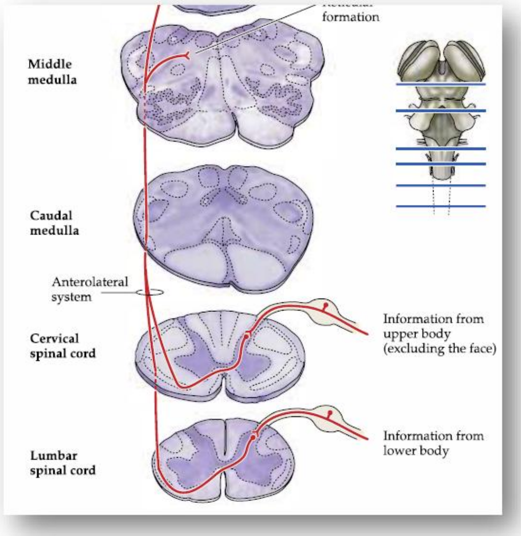
In the posterior horn of the spinal cord, you can see that the Ad fivers get to lamina 1 and 2 with the C fivers, while the tactile Ab fivers get to the lamina 4. Clearly the painful stimulation is somehow getting up to the brain also modulated by a tactile component. For Ad and C fivers that convey two stimuli, in order to reduce the pain stimulation either you provide a tactile stimulation to the side of pain or you move because you want to increase the proprioceptive stimulation, and these elements are actually gating and blocking the information that goes up to the brain.

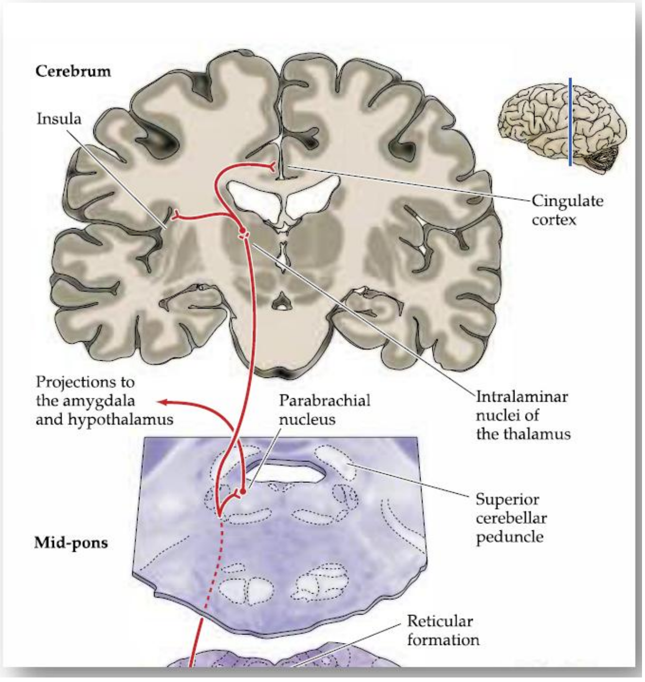
Painful stimuli are so important that the brain, from lower part of the body up to the head, has five different ascending pathways that convey information up to it. The most important one is the spinothalamic. As for painful stimulation the information enter the posterior horn and cross immediately to the contralateral part and goes up to the brain stem, and this is very important, because even if painful stimulation gets to a different pathway that ascend parallel to the posterior one for the tactile component, also this information gets immediately to the post-central gyrus, because it is important, beside this information of urgency, that the brain is aware where is the source of pain.



Some fivers of the same pathway are actually going to this specific portion of a nucleus (bundles neurons), the reticular formation, that is very important for arousal, if you have a pain stimulus you wake up, your alertness is increased, and you attention is verted towards the painful stimulus. It is the reticular formation in the brain stem that regulates this aspect.

There is also another pathway other than the spinothalamic (important to localize and characterize the source of the pain) and the spinoreticular (important to alert the brain about the painful stimulus). We have the sinomesencephalic, that is actually getting information from very deep structure that is called periaqueductal gray matter. The mesencephalo has this role of activating and regulating all the body structure, so together with the spinoreticular elements, this alertness is also reflected in several changes at the peripheral part of the body (heart rate increasing, sympathetic system increased). But more importantly this periaqueductal gray matter is a sort of an hub that modulates the incoming and outcoming painful information that goes to the brain, a gate that modulates the access of painful stimulation.

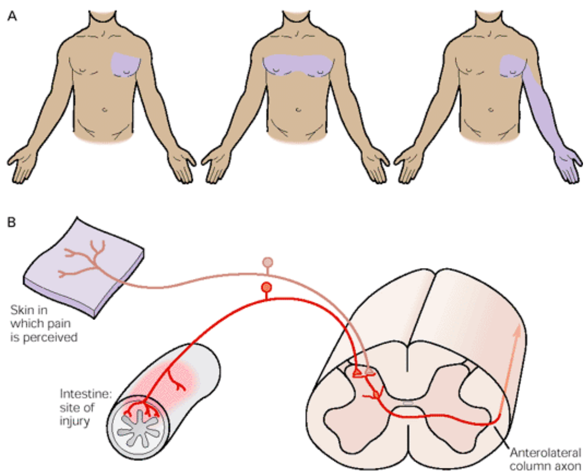




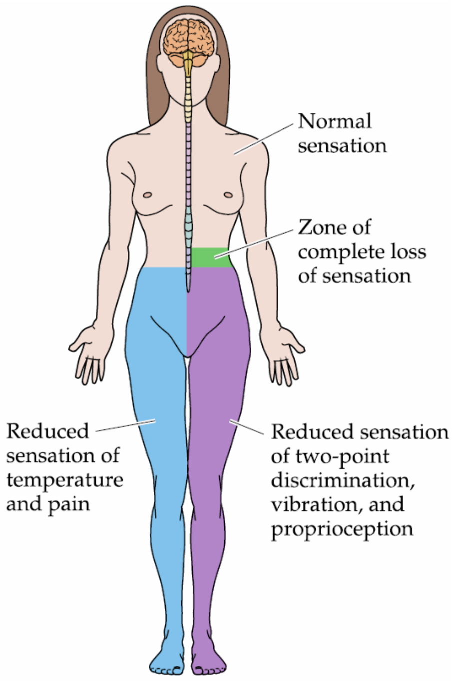
Then there are two other pathways that are primarily related to two systems getting to cervical components that may actually send information from the cervical and the face and the neck up to the brain. One feature of these elements is that they relay to the reticular formation, but then they start to get other branches of this information, to the amygdala and the hypothalamus, but also the insula (primarily considered as the central pain area) and the cingulate cortex, and this final is important because it seems to be fundamental to monitor conflict or painful situation.

This idea of having a primary area for pain stimuli has never been confirmed, and typically we describe that all brain areas involved in the process of pain have a pain matrix. Typically pain is a modality that does not rely on a primary area, it relies on more cortical areas that actually elaborate different elements of the painful information.

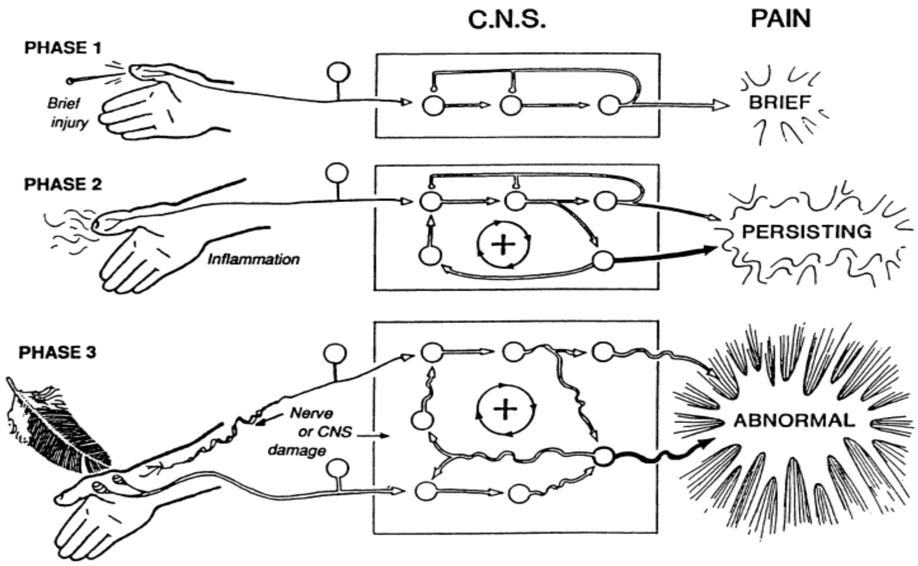
When we have afference that innervate internal organs of the body, they also have fivers that monitor painful situation and rely on the posterior horn of the spinal cord. What happen is that when information from the internal organ gets to the posterior horn, the brain is just getting information from that neuron but is not actually aware that the information is coming from the body organ or the skin. What our brain is actually getting is therefore a representation of pain on the skin surface, and this is called referred pain.



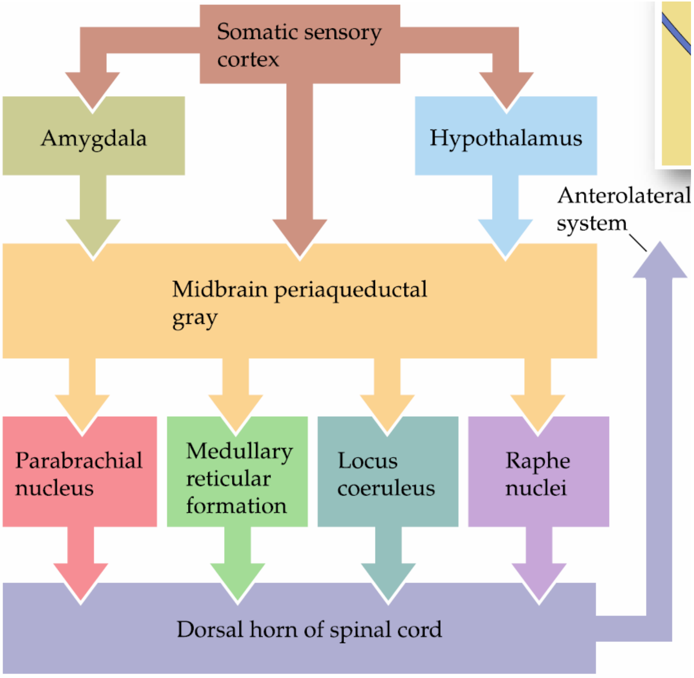
The information is going up to the brain ascending to the antero-laterally portion contralaterally, but ipsilaterally we have tactile information, so in one half of the spinal cord we are containing tactile information from one side of the body and pain information from the opposite side of the body. In those cases where we have lesions of just half of the spinal cord, you have a situation that is called dissociated sensory loss, so you may have in one side reduced sensation in touch and on the other part a reduced sensation of temperature and pain, and that is why the thermal and painful information are crossing immediately the spinal cord and the tactile information crosses at the level of the brain stem, but it ascends ipsilaterally.



Pain is a very tough process. One of the main issues that goes with pain is that typically initially you have a very brief stimuli, but most of the time the pain can persist and then have a persist painful stimulation (inflammation, formation). When it persists at the level of the periphery and the brain you have a sort of sensitization, a circuit of pain perception enhanced. You have very subtle stimuli that due to this reinforcement at the brain level you can perceive as harmless and painless.



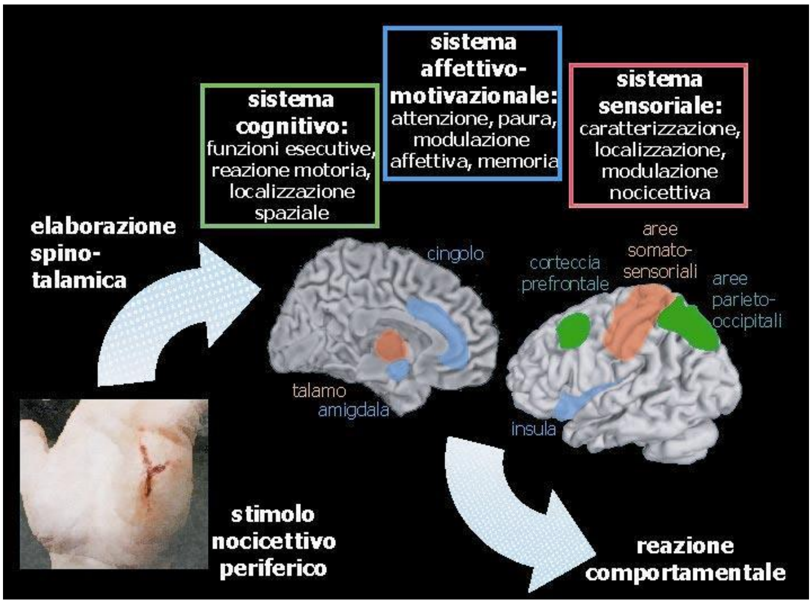
What is interesting about pain modulation is that the information doesn't just go up to the brain but is also modulating many different aspects. And this is what we call pain matrix.



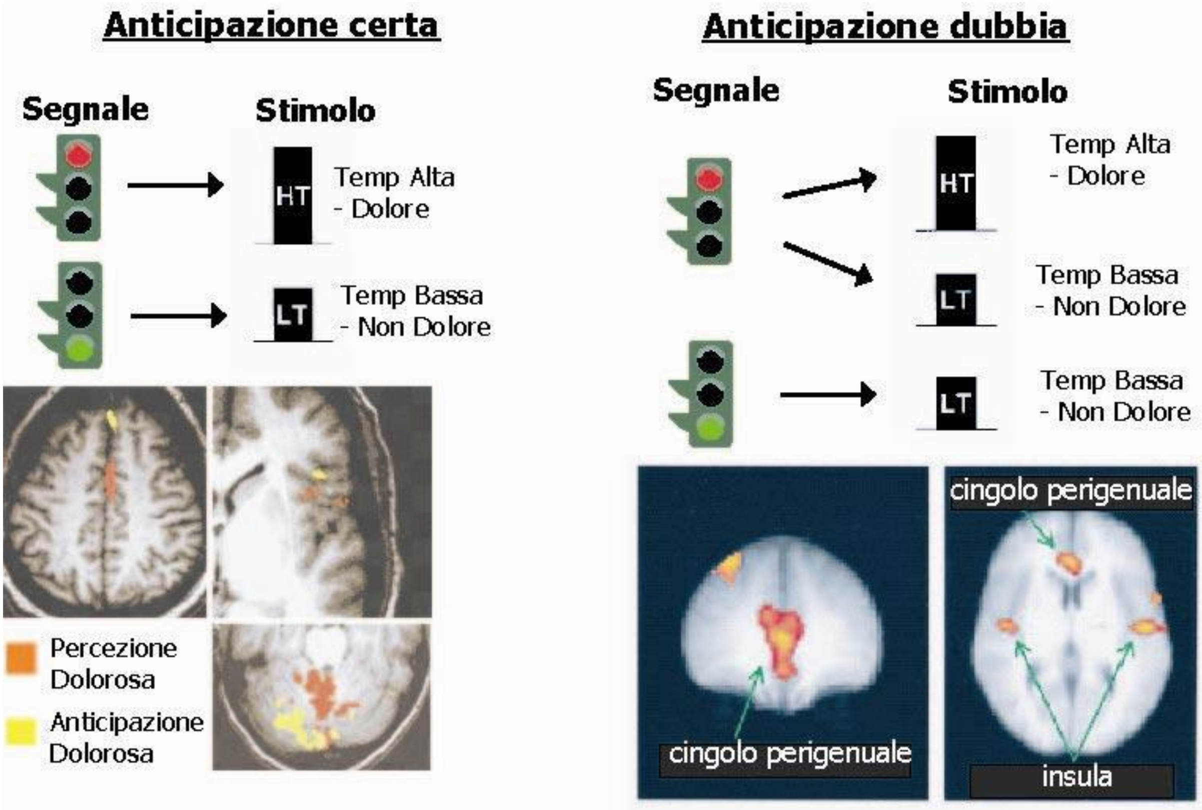
When we have a painful stimulation, we should consider three components:

- A sensory perceptive component, that primarily is useful to characterize intensity and localization of the stimulus, and this is task of the post-central gyrus.
- An affective component of pain, primarily related to the cingulate cortex, the amygdala and the insula, that modulate the affective components of pain (alertness, fear).
- A cognitive component, primarily relying in frontal-parietal regions. A primarily modulated attention, we have to localize spatially the source of the painful stimuli and act a motor reaction in response to the stimulus.

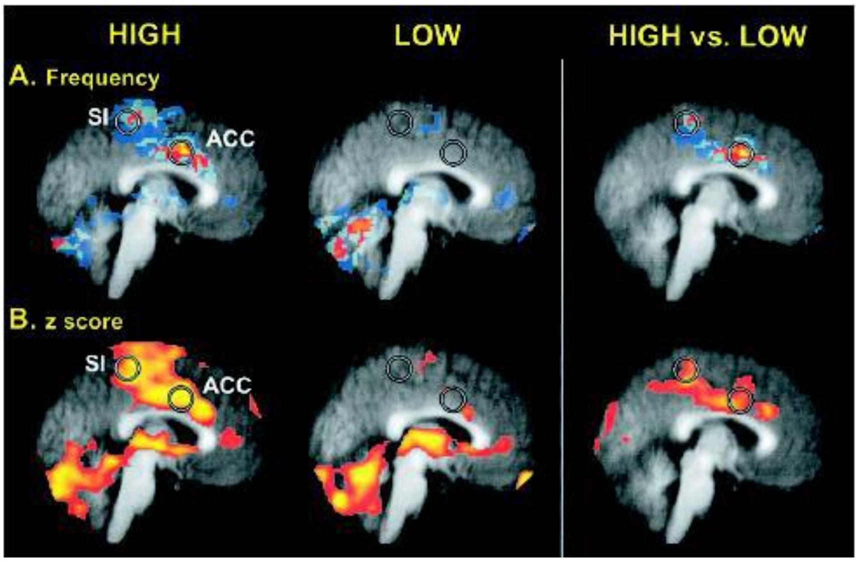
Imaging studies are also trying to visualize the antero-lateral system, so we have also functional responses in other regions.



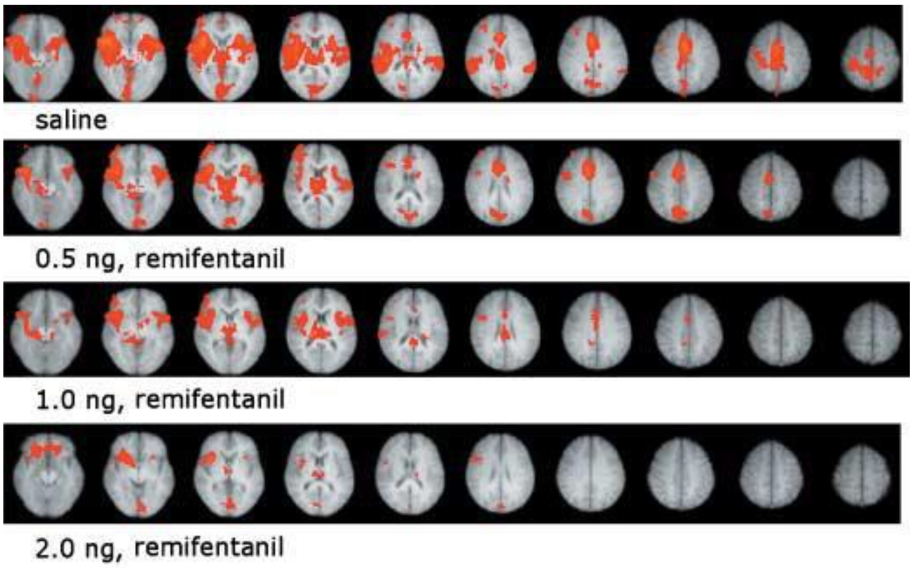
It is possible to have very simple experimental design where we play what is called anticipation of pain. This is typically used in experiments of fear conditions, you have green light that is no harmful stimulus and red light the painful stimulus. When the brain is already expecting a painful stimulus, it happens. When you anticipate pain, you have an affective component that already is having you to expect pain.



These aspects of sensitivity to pain are quite intriguing, because each of us has a different sensitivity to. If you compare high vs slow response, there is no specific difference in the post central gyrus, their true difference is in the anterior cingulate cortex. The same exact stimulus is processed at the thalamic level in the same way in people in high or low sensitivity to pain. But then you can see in the anterior cingulate that you have an exaggerated response to pain for high sensitivity to pain persons.



The higher is the concentration of analgesics the lower is the response, primarily in the anterior cingulate cortex. This is what you can study, the effects of opioids. If you give placebo, 30% of people improve.



The interesting part for us is to have a modulation of pain in the anterior cingulate. There were several examples of PET studies measuring blood flow with patients susceptible to achnosis? and through hypnosis they were having them to perceive as less or more unpleasant their handling of cold water. Only saying that they were going to get a painful stimulus, in the anterior cingulate will appear a higher response.

For most of these studies you could claim that the anterior cingulate is not really the pain processing affective component, but sort of like decisional aspect on whether to react to painful stimuli. This portion of the brain is actually getting information directly from the spinal cord, the thalamus, to modulate our behaviour on pain perception.

Vision

Here we would see a much finer degree of description and complexity of representation, as vision is the most well extended studied modality across all our sense.

The information is seen in two dimensions by the two eyes and then processed 3D mentally in our brain. There are at least two parallel and interacting pathways that process information. More than half of the cortex is devoted to visual processing. The whole occipital lobe is subserving visual processing; there is no other region that does the same for a single function.

What we actually can say is that the visual system transforms transient like patterns (even from low level features we can reconstruct the image) on the retina in to stable representation of the tri dimensional world.

The eyes adapt and change their shape to get the light getting properly to the posterior side of the eye. But when the shape of the eyes changes and the ability to modulate this accommodation is lost, we typically have disorders that need correction to improve our ability to see. The eyes have a protective surface that is mostly transparent (in the cornea), and then the rest of the eyes is the sclera and is mostly peripheral to this central transparent part. Then we have two chambers (anterior and posterior) divided by the lens, and these are filled with two aqueous solutions, the